

LEVEL 2 • 4

MISSION BRIEFING

Conjugate of a Complex Number

Every complex number has a mirror twin across the real axis. In this quest you'll learn to find it, prove its properties, and use it to conquer division problems.

Concept

Formulas

Solved Sum

Visual Quest

Boss Challenge

1

WHAT IS A CONJUGATE?

The Concept

Definition The conjugate of $x + iy$ is $x - iy$. Simply flip the sign of i .

1

Notation

The conjugate of z is written $\bar{z}$ (read "z bar").

2

Geometric meaning

$\bar{z}$ is the mirror image of z , reflected across the real axis.

3

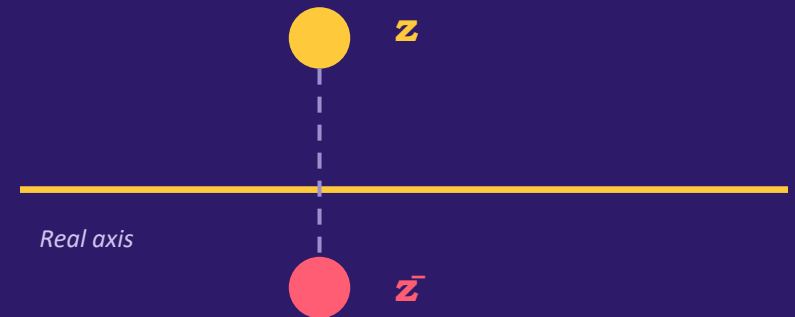
Superpower

z multiplied by $\bar{z}$ always gives a real number — useful for division!

Quick check: $z = 2 - 5i \rightarrow \bar{z} = 2 + 5i$

THINK OF IT LIKE...

A pond reflection



Same distance from the real axis — one above, one below.

2

COLLECT YOUR POWER-UPS

Properties of Conjugates

1

$$\text{conj}(z_1 + z_2) =$$

$$\bar{z}_1 + \bar{z}_2$$

2

$$\text{conj}(z_1 - z_2) =$$

$$\bar{z}_1 - \bar{z}_2$$

3

$$\text{conj}(z_1 \cdot z_2) =$$

$$\bar{z}_1 \cdot \bar{z}_2$$

4

$$\text{conj}(z_1 / z_2) =$$

$$\bar{z}_1 / \bar{z}_2 \quad (z_2 \neq 0)$$

5

$$\text{Re}(z) =$$

$$(z + \bar{z}) / 2$$

6

$$\text{Im}(z) =$$

$$(z - \bar{z}) / 2i$$

7

$$\text{conj}(z^n) =$$

$$(\bar{z})^n$$

8

$$z \text{ is real } \Leftrightarrow$$

$$z = \bar{z}$$

9

$$z \text{ is imaginary } \Leftrightarrow$$

$$z = -\bar{z}$$

10

$$\text{conj}(\bar{z}) =$$

$$z$$

Tip: properties 1–4 and 7 hold for sums, products, quotients and powers — conjugation "distributes" through every basic operation.

3**SOLVE THE QUEST**

Worked Example

Quest: For $z = 4 + 7i$, find $\bar{z}$ and verify that $\operatorname{Re}(z) = (z + \bar{z}) / 2$.

1**Step 1 — Find the conjugate**

Flip the sign of i : $z = 4 + 7i \rightarrow \bar{z} = 4 - 7i$

2**Step 2 — Add z and $\bar{z}$**

$$z + \bar{z} = (4 + 7i) + (4 - 7i) = 8$$

3**Step 3 — Divide by 2 and compare**

$$(z + \bar{z})/2 = 8/2 = 4 = \operatorname{Re}(z) \quad \checkmark \text{ Verified!}$$

**QUEST COMPLETE!****+10 XP**

$$\bar{z} = 4 - 7i$$

Re(z) confirmed equal on both sides

4

ENTER THE MIRROR REALM

Geometric Picture

1

Plot $z = 4 + 7i$ as an arrow from the origin.

2

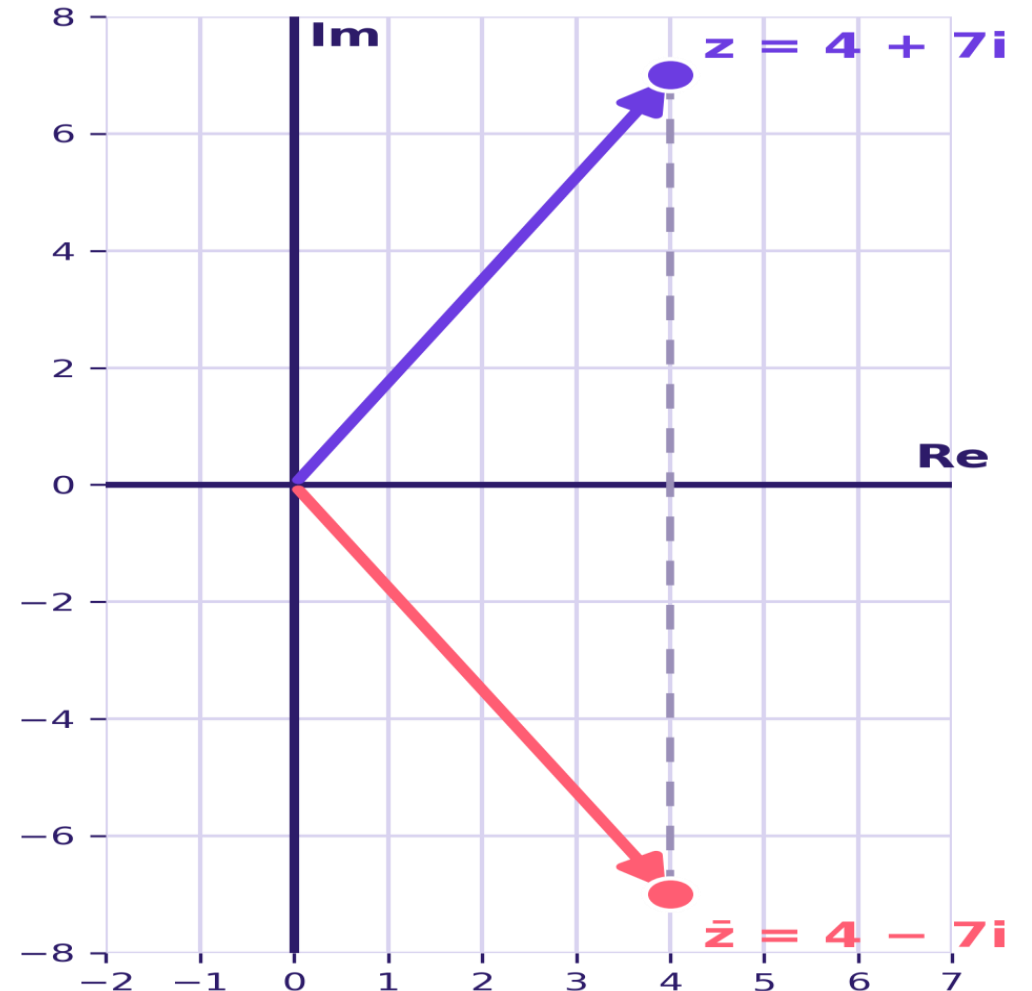
Its conjugate $\bar{z} = 4 - 7i$ sits the same distance below the real axis.

3

The real axis acts like a mirror — reflecting z straight down to $\bar{z}$.

4

Both points always share the same real part.



LEVEL COMPLETE

Recap & Boss Challenge

Definition

Flip the sign of i : $x+iy \rightarrow x-iy$

Geometry

$\bar{z}$ is the reflection of z across the real axis

10 Properties

Conjugation distributes over $+$, $-$, $\times$, $\div$, powers

BOSS CHALLENGE

Find $\bar{z}$ for $z = -6 + 2i$, then check whether z is purely imaginary.

Hint: flip the sign of i for the conjugate, then use property 9 — z is purely imaginary if and only if $z = -\bar{z}$.